Curl and Divergence

$= \langle P, Q, R \rangle$

Goal:

To calculate the curl and divergence of a vector field $\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z)\rangle$.

$f(x,y,z)$

$\left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle f(x,y,z)$

Definitions:

1. The Del Operator: $\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$

Gradient: $\nabla f = \langle f_x, f_y, f_z \rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$

2. Divergence of **F**:

$$\operatorname{Div} \vec{F} = \nabla \cdot \vec{F} \overset{\text{def}}{=} \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle P, Q, R \rangle$$

$$= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

3. Curl of **F**:

$$\operatorname{Curl} \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Example:

Let $\mathbf{F} = \langle \overset{P}{xe^{-y}}, \overset{Q}{xz}, \overset{R}{ze^{y}} \rangle$. Find the divergence and curl of **F**.

$$\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = e^{-y} + 0 + e^{y} = \boxed{e^{-y} + e^{y}}$$

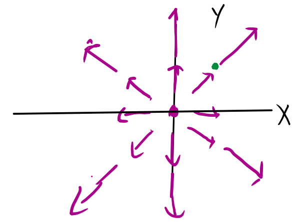
$$\operatorname{curl} \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xe^{-y} & xz & ze^{y} \end{vmatrix} = (ze^{y} - x)\vec{i} - (0 - 0)\vec{j} + (z - (-xe^{-y}))\vec{k}$$

$$= \boxed{(ze^{y} - x)\vec{i} + (z + xe^{-y})\vec{k}}$$

The divergence measures how much "stuff" is emanating from a point in $\vec{F}$.

Ex $\quad \vec{F} = x\vec{i} + y\vec{j}$

$\text{div}\,\vec{F} = 1 + 1 = 2$



lots of divergence at $(0,0)$ for example

ex $\quad$ Curl $\vec{F}$ gives an idea of how much $\vec{F}$ rotates at a point. $\quad \boxed{\vec{F} = -y\vec{i} + x\vec{j}}$ $+0\vec{k}$ $\leftarrow$ represents circular motion (lots of curl)

$f_x \; f_y \; f_z$ when $\vec{F}$ conservative

$F = \langle P, Q, R \rangle$

Theorem:

F is conservative if and only if $\underline{\quad \text{curl}\,\vec{F} = \vec{0} \quad}$

Proof:

($\Rightarrow$) Assume $\vec{F}$ is conservative. $\quad \text{Curl}\,\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_x & f_y & f_z \end{vmatrix}$

$= (\;0\;)\vec{i} - (\;0\;)\vec{j} + (\underbrace{f_{yx} - f_{xy}}_{=0})\vec{k}$

since mixed partials are equal.

($\Leftarrow$) Assume curl $\vec{F} = \vec{0}$. Showing $\vec{F}$ coservative is harder and requires Stokes' Th. (16.8)

Example:

Determine whether $\mathbf{F}(x, y, z) = \underbrace{e^z}_{f_x}\mathbf{i} + \underbrace{1}_{f_y}\mathbf{j} + \underbrace{xe^z}_{f_z}\mathbf{k}$ is conservative. If so, find a $\underbrace{\text{potential function for}}_{\nabla f = \vec{F}}$ **F**.

$\nabla f = \vec{F}$ $\;\llcorner$ potential fctn for $\vec{F}$.

exercise: $\quad$ Show $\;\text{Curl}\,\vec{F} = \vec{0}$ ✓

$f(x,y,z) = \int e^z\,dx = xe^z + \overbrace{g(y,z)}^{\text{constant w.r.t } x} \leftarrow$

$f_y = g_y(y,z) = 1$

$\int g_y(y,z)\,dy = \int 1\,dy$

$g(y,z) = y + h(z) \leftarrow$

$f(x,y,z) = xe^z + y + h(z)$

$\rightarrow f_z = xe^z + 0 + h'(z) = xe^z$

$\cancel{x}e^z + h'(z) = x\cancel{e^z}$

$h'(z) = 0$

$h(z) = c$, constant

$\boxed{f(x,y,z) = xe^z + y + c}$